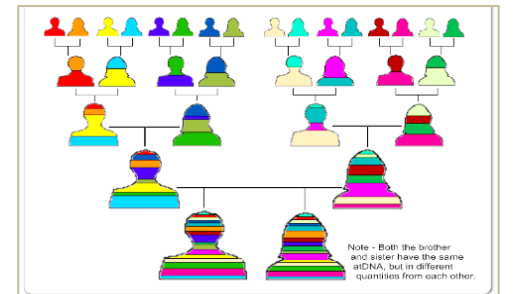




FORENSIC DNA PROFILING

Dinesh K. Jha
Chief Scientific Officer



Key significances of forensic DNA: lead, rigor and proof

INVESTIGATIVE

Occurrence (e.g., touch DNA)

SCIENTIFIC

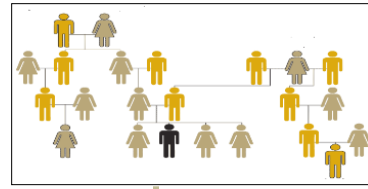
Meets the highest scientific/legal standards -
Frye test, Daubert and
NAS (2009)

LEGAL

Individuality - Provides
definitive inclusion or
exclusion.

Individual ID

Biological relationship



Application in
Human & Non-human forensics



Species ID
Geographical origin
Sex ID

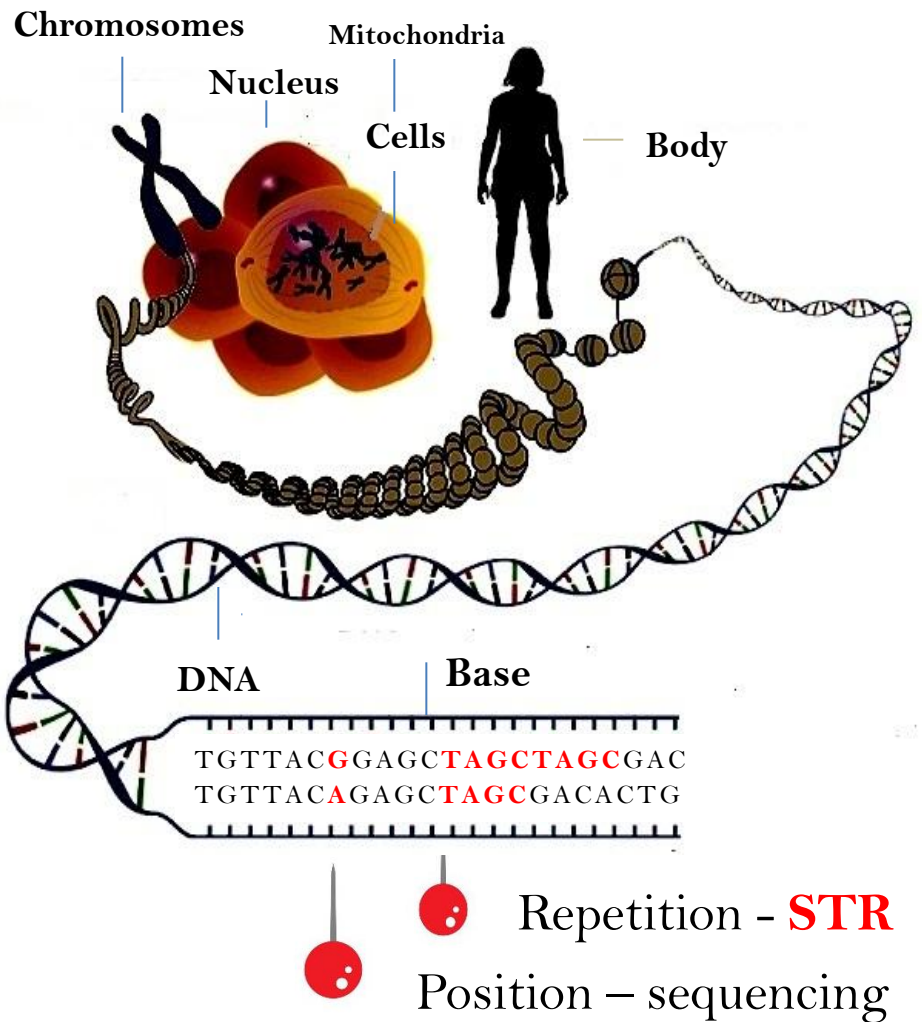
investigative lead tool



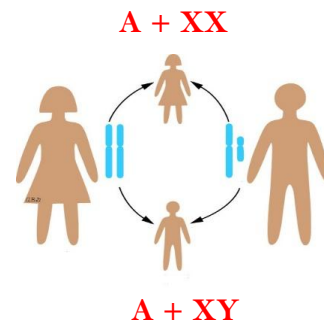
DNA database

Pinpoint Ancestry/caste Sketch

Major references : *The National criminal procedure code 2017; 21 (1), Kasurko anusandhan sambandhi niyamabali, 2075, 24 (2); The Enforced Disappearances Enquiry, Truth and Reconciliation Commission Act, 2071, 14 (7), Pahichaan nabheyka tatha bebarise las bybasthapan nirdeshika, 2078, 5 and 6. The National civil code 2017 ; 94/95, 99, 107, The national criminal code 2017; 189 (Ga). Abortion. National parks and wildlife conservation ACT 1973; 26. CITES ACT, 2017*



DNA : DEOXYRIBO NUCLEIC ACID



Autosomal - chromosome
Y - chromosome
X - Chromosome

Types	Inheritance	Application
Autosomal STR	Both parents	Most biological relationship, ID
Y-STR	Father-son	Male ancestry, rape
X-STR	Mother-both Father-daughter	Siblings.
mt DNA sequencing	Mother - both	Female ancestry, degraded sample
SNPs	Highly degraded sample, ultimately used.	

- NAFOL, Khumaltar : All facilities
- NPCFSL, Samakhushi : Autosomal and Y STR
- NAFOL, Butwal & Dhangadhi : Autosomal STR and sequencing
- NAFOL, Surkhet and NPCFSL, Pokhara : Set up phase.

Workflow



Sample & sampling	DNA extraction & amplification	DNA profile generation	Comparison	Reporting
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MATCH – statistical analysis

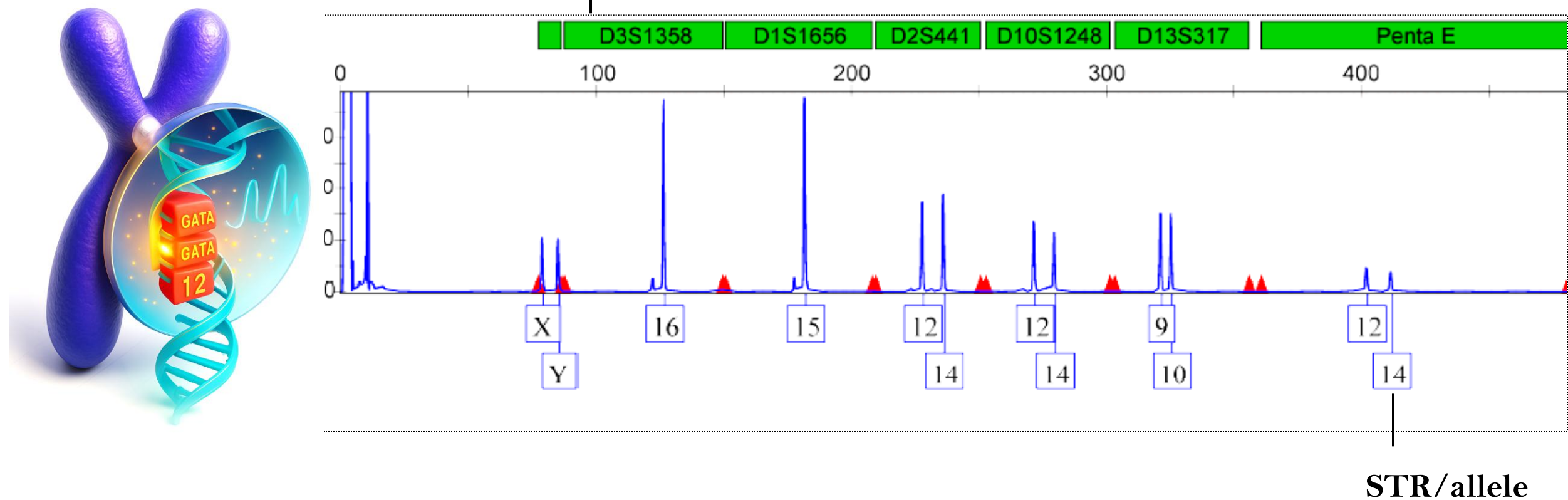
MISMATCH

INCONCLUSIVE

In forensic DNA profiling, the evidentiary value depends on proper sampling (what, whose, how), an unbroken chain of custody, validated laboratory methods/equipment, and scientifically sound interpretation.

Electropherogram : DNA analyzer generated DNA profiles

Locus/marker : Physical location of interested STR in chromosome



The FBI recommends **20 core DNA loci** for forensic identification in CODIS, ensuring extremely high accuracy with random match probabilities often below 1 in a trillion.

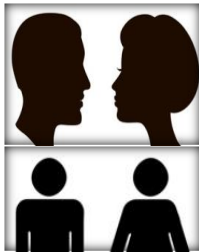
Identification of an individual relies on analyzing genetic similarities in cells or patterns of inheritance to establish unique biological profiles.



Direct ID : Own biological sample



Indirect ID : Close biological relative/s



Biological relationship analysis: Establishes genetic links through inheritance patterns, with highest accuracy in trio paternity, lower in duo paternity, and further reduced in sibling and extended kinship analyses, depending on method and reference samples.



Paternity

Autosomal STR

Maternity

Autosomal STR



Y STR

Mt DNA



X STR

Mt DNA



-

Mt DNA

Locus	Mother	child	Al. father
FGA	6, 8	6, 4	4, 9

Result interpretation, opinion & reporting

Two or more mismatches (or three in conservative labs) exclude an individual, while a match indicates inclusion expressed as a probability, with paternity >99.9% (K. Hummel standard) considered practically proven

Cases	Opinion
Paternity	
Exclusion	Person is excluded as the biological father
Inclusion	Concluded as the biological father (≥ 99.9 based)
Male ancestor	Found to same (or different) male lineage
Identification - Inclusion	
indirect	Combined Random parents not excluded is 0.0005 etc
direct	Random match probability is 0.0002 etc



1. **DNA is a highly reliable tool** for establishing biological relationships and criminal identification, surpassing other scientific and non-scientific evidence *Roshan Maharjan vs. Siddhilal Maharjan* (D.N 7784), *Sitadevi Thakur vs. Budhani Thakur* (D.N 8501), *Rajiv Gurung vs. Neeta Gurung* (D.N. 8578), *Badriprasad Prasai vs. Radhika Majhi* (D.N. 9294), *Jogendra Ahir vs. Rabadi Ahirni* (D.N. 9395).
2. **Subject of court**, denies, if conclusion can be drawn from other evidences. *Buddhu Ahir vs. Jang bdr Ahir* (D.N 8250), *Tek B. Thapa vs. Indira Adhikari* (D.N. 8272), *Lakshmi Shrestha vs. Makhmali Mishra* (D.N. 9724)
3. **Admissible** evidence. (*Makhmali Mishra vs. Laxmi Shrestha*, D.N. 8483).
4. **Inadmissible** due to procedural and sampling errors. *Rakesh K Singh vs. Himali gole/GoN* (D.N. 7810), *Rajiv Gurung vs. Neeta Gurung* (D.N. 8578), *Badriprasad Prasai vs. Radhika Majhi* (D.N 9294).
5. **DNA database, DNA ACT**. *Rajiv Gurung vs. Neeta Gurung* (D.N 8578), *Ramesh Parajuli vs. PM* (D.N. 9732).
6. **Not a single/conclusive evidence** to be considered. *Prakash Raj vs. Pradeep Raj* (C.N. 072-CI-1935)

Supreme court view

In Pradip KC v. Salina Malla (C.N. 076-CI-0489, 2081.08.16) : DNA profiling is the most reliable scientific and decisive evidence for paternity, based on repeated DNA results, conclusively excluded the alleged father based on DNA results, overturning the lower courts' decisions

!!!! Bayesian method: Integrates prior (non-genetic) evidence with DNA results to calculate posterior odds for judicial evaluation.

Commonly raised quesstions

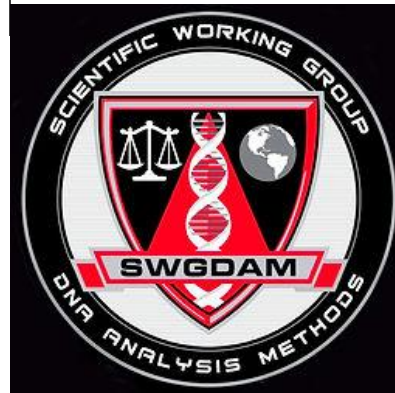
Sample and sampling

- Witness/muchulka
- Necessary photography
- Documentation

Delay examination

- May partial but NOT false results

Reliability of results



- Forensically validated tools
- Reproducibility of results

RE Verification Examination

- Prolonged storage of sample
- Stored all genuine data
- **Re-examination** in practices



सर्वोच्च अदालत

(अनुसन्धान तथा योजना महाशाखा)



रामशाहपथ, काठमाडौं

फोन : ४२००७२९, ४२००७२८

४२००७५०

फ्याक्स : ४२००७४९

पो.ब.नं. २०४३८

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पत्र संख्या : ०८७/०८८

प्राप्त पत्र संख्या र मिति :

ब.नं. ३८८

श्री शिक्षा, विज्ञान तथा प्रविधि मन्त्रालय

सिंहदरबार, काठमाडौं ।

मिति: २०८१/०८/०७

विषय: निर्णयको जानकारी सम्बन्धमा ।

उपरोक्त विषयमा विभिन्न अदालतमा सामो समयदेखि विचाराधीन मुद्दा सम्बद्ध नमूनाको विधि विज्ञान प्रयोगशालामा परीक्षण भई सोको प्रतिवेदन प्राप्तमा भएको दिलाई समेतका सन्दर्भमा सर्वोच्च अदालतको अनुसन्धान तथा योजना समितिका अध्यक्ष एवं सर्वोच्च अदालतका वरिष्ठतम् माननीय न्यायाधीश श्री सपना प्रधान मल्लज्यूको अध्यक्षतामा सो समितिका सदस्यहरु एवं तहाँ मन्त्रालयका सचिवज्यू (विज्ञान तथा प्रविधि), प्रहरी प्रधान कार्यालय अपराध अनुसन्धान विभागका प्रमुखज्यू, राष्ट्रिय विधि विज्ञान प्रयोगशाला तथा नेपाल प्रहरी विधि विज्ञान प्रयोगशालाका पदाधिकारीहरु समेतको उपस्थितिमा मिति २०८१ साल मंसिर ४ गते सर्वोच्च अदालतमा बसेको बैठकले गरेका निर्णयहरु आवश्यक कार्यार्थ निर्णयको प्रतिलिपी यसैसाथ राखी पठाइएको व्यहोरा अनुरोध छ ।

बासुराम दाहाल

सहरीजिष्टार

बोधार्थका लागि कृपया:

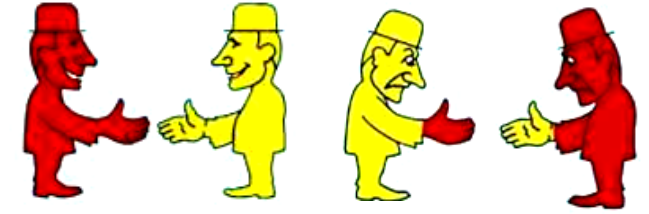
श्री प्रहरी प्रधान कार्यालय, नक्साल, काठमाडौं ।

✓ श्री राष्ट्रिय विधि विज्ञान प्रयोगशाला खुमलटार सलिलपुर ।

श्री नेपाल प्रहरी विधि विज्ञान प्रयोगशाला, सामाखुसी काठमाडौं ।

Challenges

- **Motherless paternity** : Increases the risk of false inclusion
- **Prenatal paternity sample** : Requires proper samples like Amniotic fluid. (ref letter. PHQ, OAG 077/78 No. 1189, 2078-01-19)
- **Sampling (contamination) issues** : especially in trace/sexual assault case samples.
- **Delay report and over pending cases.** (~1800 DNA cases pending in NPCFSL)



Briefly

- Forensic DNA is a vital and globally accepted scientific tool.
- International platforms and facilities for DNA analysis are widely available.
- In Nepal, criminal identification use is limited and is mandatory mainly in positive semen cases (*Sandip Nepali et al., D.N. 10148*).
- Annual casework covers individual identification (~ 500), biological relationships (~ 200), and species identification (~ 150).
- Key challenges exist at pre-technical (sampling), technical (methods and specialized units, especially in sexual assault), and post-technical (judicial admissibility) stages.
- DNA is a powerful tool, but its value depends on validated methods, proper interpretation, and legal context, **and it must not be treated as automatically conclusive.**